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Emerging threats of high biofilm formation and antibiotic resistance in clinical methicillin-resistant *Staphylococcus aureus* (MRSA) isolates from Pakistan

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ABSTRACT

Objectives: This multicenter study, conducted from a One Health perspective, aimed to comprehensively examine the prevalence of methicillin-resistant *Staphylococcus aureus* (MRSA) infections and their biofilm-forming capabilities in Pakistan. Phylogenetic analysis of the study isolates was also performed.
Methods: A total of 150 MRSA isolates were screened from various clinical samples using Cefoxitin antibiotic discs. Genotypic confirmation was conducted through *mecA*, *S. aureus*-specific *nuc*, and 16S rRNA genes. Biofilm formation was assessed using Congo red agar (CRA) and slime layer quantification methods. The intercellular adhesion (*ica*) operon genes, specifically *icaA* and *icaD*, were detected. Phylogenetic analysis utilized the 16S rRNA sequences. Statistical associations between various parameters were determined using chi-square analysis.
Results: The presence of the *mecA* gene was observed in 131 out of 150 isolates (87.3%). CRA identified 28% and 40% of isolates as strong and moderate biofilm producers, respectively, while 9.3% were classified as non-biofilm producers. The slime layer assay exhibited higher sensitivity, classifying only 4.7% of isolates as non-biofilm producers. Biofilm-forming genes *icaA* and *icaD* were detected in 85.3% and 86.7% of the isolates, respectively. Antibiotic resistance was more prevalent among biofilm-forming isolates, particularly against ciprofloxacin, levofloxacin, erythromycin, trimethoprim-sulfamethoxazole, and fosfomycin. Ceftaroline demonstrated efficacy irrespective of biofilm-forming abilities. Conversely, non-biofilm producers exhibited complete susceptibility to clarithromycin and tigecycline.
Conclusions: Clinical MRSA strains exhibit a substantial potential for biofilm formation, contributing to a resistant phenotype. Routine antibiotic testing in clinical settings that overlook the biofilm aspect may lead to the failure of empiric antibiotic therapy.

1. Introduction

Staphylococcus aureus, a significant pathogen in both hospital and community settings, ranks as the second most prevalent cause of nosocomial infections. In Pakistan, Methicillin-resistant *S. aureus* (MRSA) is the second major contributor to device-related infections following *Escherichia coli*, constituting 15% of total infections (Cabrera et al., 2020; Sohail and Latif, 2017). Particularly concerning is the frequent detection in skin infections, which can serve as a potential source for life-threatening bacteremia. The establishment of biofilm during skin infections further accentuates the risk of bacteremia (Tong et al., 2015).

Biofilm, a consortium of microbiological communities, is fortified by an extracellular matrix comprising water, extracellular DNA (eDNA), proteins, nutrients, and polysaccharides. The initial stage of biofilm formation involves bacterial adhesion to the host surface, primarily facilitated by polysaccharide intercellular adhesin (*ica*), also known as poly-*N*-acetylglucosamine, encoded by the *icaADBC* operon. Additionally, *ica*-independent mechanisms, such as fibronectin-binding proteins and major autolysin, contribute to biofilm formation, albeit with lower virulence. *S. aureus* tends to form biofilms on medical device surfaces, rendering eradication of infections challenging. These strains exhibit high resistance to antibiotics and host immune defences, leading to persistent infections (Tong et al., 2015; Pozzi et al., 2012). Recognizing the escalating threat of MRSA resistance, the World Health Organization

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(WHO) designated MRSA as a high-priority pathogen in 2017 (Tacconelli et al., 2018).

In developing nations like Pakistan, the incidence of MRSA infections is on the rise (Bilal et al., 2021). Previous studies on biofilm-forming aspects in Pakistan are limited and have various shortcomings, such as small sample sizes, localized sampling, or focus on specific types of infections (Sohail and Latif, 2018; Awan et al., 2022; Rahim et al., 2016). Furthermore, the correlation between MRSA biofilm formation and antibiotic resistance remains less explored. This study aims to fill these gaps by analysing biofilm formation in clinical MRSA isolates across multiple healthcare centres and assessing its association with antibiotic resistance. This study also explored the phylogenetic distribution of the study isolates in conjunction with the previously reported isolates in Pakistan and neighbour countries.

2. Methods

2.1. Bacterial isolates

This study spanned from September 2018 to August 2019 and included all the MRSA isolates over the study duration. A total of 150 methicillin-resistant *Staphylococcus aureus* (MRSA) isolates were systematically retrieved from different patients (visiting or admitted) having certain infections. To ensure the generalizability of findings across the study region, this study encompassed diverse tertiary healthcare settings including Jinnah Hospital Lahore (JHL), Lahore General Hospital (LGH), Services Institute of Medical Sciences (SIMS), and CitiLab & Research Centre Lahore (CRC). Non-duplicate MRSA isolates were exclusively included, while methicillin-sensitive isolates were deliberately excluded.

2.2. Characterisation of bacterial isolates and antibiogram profile

Phenotypic identification of MRSA isolates was conducted using the characteristic growth patterns on mannitol salt agar, blood agar, and DNase agar. Methicillin resistance was determined using Cefoxitin antibiotic discs. The antimicrobial susceptibility testing was done using the disc diffusion method following the Clinical and Laboratory Standards Institute (CLSI) established guidelines (*Performance Standards for Antimicrobial Susceptibility Testing*, 2019). A panel of 18 antibiotics (Bioanalyse®, Turkey) was employed, including cefoxitin (FOX-30 μg), clindamycin (DA-2) μg, erythromycin (E-15 μg), clarithromycin (CLR-15 μg), linezolid (LZD-30 μg), ciprofloxacin (CIP-5 μg), levofloxacin (LEV-5 μg), gentamicin (CN-10 μg), tetracycline (TE-30 μg), doxycycline (DO-30 μg), trimethoprim-sulfamethoxazole (SXT-25 μg), nitrofurantoin (FD-300 μg), chloramphenicol (C-30 μg), ceftaroline (CPT-30 μg), fosfomycin (FOS-50 μg), fusidic acid (FA-10 μg), tigecycline (TGC-15 μg), and Quinupristin-Dalfopristin (QD-15 μg). FA and TGC were interpreted according to the European Committee on Antimicrobial Sensitivity Testing (EUCAST) established breakpoints version 10.0 (European Committee on Antimicrobial Susceptibility Testing, 2020). FOS-50 sensitivity interpretation followed the Antibiotic Committee of the French Society of Microbiology (CA-SFM) 2019 recommendations (*Comité de l'antibiogramme de la Société Française de Microbiologie: recommandations* 2019 V.1.0, 2019).

2.3. Biofilm detection and quantification

Biofilm formation was assessed using Congo red agar (CRA) (Freeman et al., 1989) and quantified through a crystal violet-based slime production assay (Stepanovic et al., 2007). Biofilm quantification assay was repeated thrice for each isolate and the mean value was used in further analysis.

2.4. Molecular characterisation of MRSA and analysis of ica operon genes

Molecular identification of *S. aureus* was achieved by amplifying Staph-specific 16S rRNA and *S. aureus*-specific *nuc* genes (Monday and Bohach, 1999; Brakstad et al., 1992). Methicillin resistance was confirmed through *mecA* and *mecC* genes using a duplex polymerase chain reaction (PCR) protocol (Khairalla et al., 2017). The molecular mechanism of biofilm formation was explored by detecting *ica* operon genes (*icaA* and *icaD*) (Arciola et al., 2001). The PCR reaction was carried out using 12.5 μl of PCR Buffer (DreamTaq Master Mix, Thermo Scientific, USA), 5 μl of template, 1 μl of each 20 μM primer and nuclease-free water up to the final reaction volume of 25 μl. The PCR conditions included initial denaturation at 95 °C for 10 min, 35 cycles of denaturation for 30 s at 95 °C, annealing for 30 s (at 51 °C for 16S-rRNA, 55 °C for *nuc*, 52 °C for *mecA* and *mecC*, and 55.5 °C for *icaA* and *icaD*), extension for 1 min at 72 °C, and final extension at 72 °C for 10 min. The reaction was carried out in a T100 thermal cycler (Bio-Rad, USA).

2.5. 16S rRNA sequencing and phylogenetic analysis

All the 150 MRSA isolates were grouped into fifteen clones based on the antibiogram profile, and different phenotypic and genotyping characters (including source of infection, biofilm forming phenotype, carriage of *mecA, icaA*, and *icaD* genes). Fifteen representative isolates were further selected for 16S rRNA sequencing. Genomic DNA was extracted using GeneJet Genomic DNA Purification Kit K0721 (Thermo Scientific, Massachusetts, USA) and sequenced from 1st Base-Asia company (Singapore). The sequences were analysed through Chromas, aligned by the CLUSTAL-W and MEGA-XI was utilized to generate the phylogenetic tree by the maximum likelihood method using the bootstrap method with a value of 1000. All the sequences were submitted to GenBank through the BankIt tool, and the accession numbers were acquired.

2.6. Statistical analysis

Statistical analysis was performed using GraphPad Prism version 8. Chi-square analysis was employed to determine the association between different categorical parameters, with a significance threshold set at $P < 0.05$.

3. Results

3.1. Demographics

The study included 150 MRSA isolates, with the majority sourced from CitiLab & Research Centre Lahore (CRC) (84 samples). Other contributions came from Lahore General Hospital (LGH) (22 isolates), Jinnah Hospital Lahore (JHL) (35 isolates), and SHL healthcare settings (9 isolates). Pus specimens constituted the predominant sample type (122/150, 81.3%), followed by ear swabs (4/150, 2.7%), and various other sources such as body fluids, urine, sputum, tracheal aspirate, blood, cerebrospinal fluid (CSF), and joint fluid. MRSA infection was recorded more in males (91/150, 60.7%) than in females (59/150, 39.3%).

3.2. Characterisation and antibiogram of MRSA isolates

All the isolates were identified as MRSA through cefoxitin resistance, *nuc*, and 16S rRNA testing. The *mecA* detection rate was 87.3% (131/150), and no *mecC*-carrying isolate was found. The highest resistance was observed against CIP at 92.7%, followed by SXT at 69.3%. LZD resistance was 8.7%, while TGC showed the least resistance at 0.7%. CPT and QD demonstrated complete susceptibility.

3.3. Biofilm quantification

Biofilm formation, assessed using Congo red agar (CRA) and slime layer production (or tissue culture plate method, TCP), exhibited variations (Fig. 1). CRA missed some biofilm-producing isolates (7/150, 4.7%). Strong and weak biofilm producers accounted for 28% and 22.7%, respectively, as detected by CRA. The slime layer assay identified 28% as strong and 28.6% as weak biofilm producers. Both methods categorized six isolates similarly, while 84 isolates were classified differently. Twenty-four isolates were identified as strong biofilm producers by both tests, and 26 as moderate. Both phenotypic methods accurately identified strong biofilm producers, with similar rates for moderate biofilm producers (40% by CRA and 38.7% by TCP).

3.4. icaA and icaD prevalence

The *icaD* gene was present in 130/150 (86.7%) isolates, while 128/150 (85.3%) harboured the *icaA* gene. One hundred and twenty-four (82.7%) isolates co-harboured *icaAD* genes. Overall, biofilm producers were 92% (138/150), i.e., they possessed either of the *ica* genes.

3.5. Association of antibiotic resistance and biofilm formation

All the biofilm-producing isolates exhibited complete resistance to CLR and TGC, and high resistance rates for SXT (97.4%), LEV (97.2%), CIP (97.1%) and E (96.9%) were observed. Fosfomycin-resistant isolates accounted for 84%, while LZD showed 88.2% resistance in biofilm-producing isolates. Biofilm formation was statistically associated with resistant phenotypes to erythromycin ($P = 0.02$), ciprofloxacin ($P < 0.01$), levofloxacin ($P < 0.01$), co-trimoxazole ($P < 0.01$), and fosfomycin ($P < 0.01$) (Fig. 2).

3.6. Phylogenetic analysis

The phylogenetic analysis revealed that the isolates clustered into two different groups. Most query sequences ($n = 9/15$) were closely related. The phylogenetic tree (Fig. 3) illustrated widespread disseminated in the healthcare settings, causing various infections across genders, irrespective of biofilm phenotype, sample source and carriage of *ica* genes. Accession numbers for the isolates were obtained from GenBank (accession nos. **MZ960140** and **OK090920** to **OK090933**).

4. Discussion

This study investigated into the significance of the biofilm-forming potential of MRSA isolates from clinical sources in Pakistan. Biofilm production was assessed using CRA, slime quantification, and the detection of *ica* genes. Phylogenetic analysis was performed on some representative isolates and comparison was also made to previously reported isolates from Pakistan and neighbour countries.

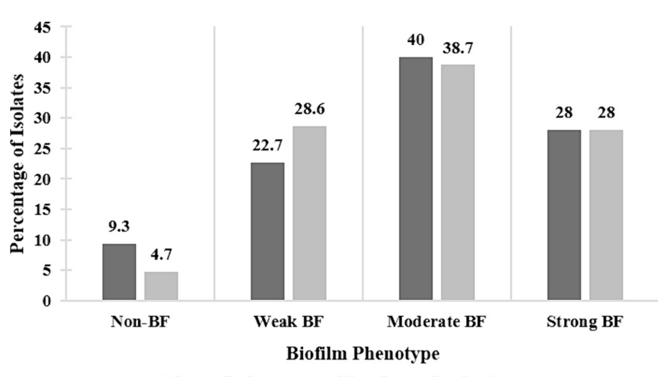

Fig. 1. Comparison of biofilm production by Congo red agar (CRA) and slime layer production assay. BF means biofilm-forming isolates. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

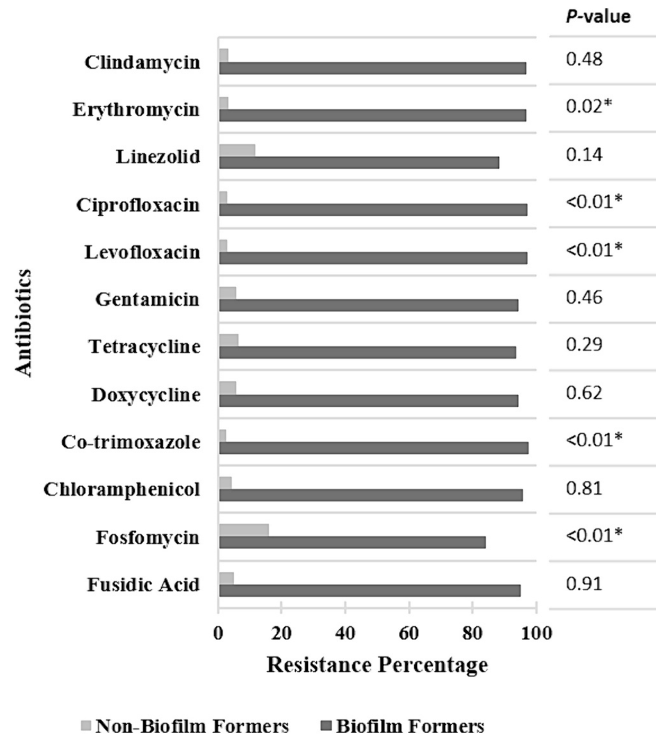

Fig. 2. Distribution of biofilm phenotype in antibiotic-resistant MRSA isolates. Statistically significant associations are shown as an asterisk (*) to the *P* value mentioned against the respective bar graph.

The study revealed a noteworthy prevalence of biofilm-forming strains, establishing findings from diverse global regions. Previous studies have demonstrated a low to moderate detection rate of 22.7% to 63.3% by slime layer production assay (Samadi et al., 2017; Vergara et al., 2017). Other studies have reported a high prevalence of 100% in different regions across the globe (Bernier-Lachance et al., 2020; Neopane et al., 2018), corresponding to the high prevalence of biofilm-forming strains in this study. The detection rate in this study, especially using the gold standard slime layer production assay (Neopane et al., 2018; Mishra et al., 2015), surpasses some previous studies which emphasize the importance of considering *ica*-independent mechanisms in MRSA biofilm formation (Tong et al., 2015).

The prevalence of moderate biofilm producers in this study, as determined by both CRA (40%) and slime layer production (38.7%) methods, is aligning to a previous study (47.9%) from Poland (Piechota et al., 2018). Strong biofilm formation was evident in a limited number of isolates, unlike findings from Iran (52.27%) (Tahmasebi et al., 2020) and China showing a very high rate (83.3%) (Yang et al., 2017). Notably, livestock-associated MRSA (LA-MRSA) and MRSA from birds and diary have been identified as strong biofilm producers (Silva et al., 2020; Thiran et al., 2018), emphasizing potential zoonotic transmission risks. The evolving epidemiology of MRSA underscores the need for vigilance in monitoring biofilm-producing MRSA strains. These differences might be attributed to the different prevailing strains globally.

Analysing the origin of isolates reveals distinct biofilm production patterns. For instance, strains isolated from urine exhibited strong biofilm production. Another study also reported significantly strong biofilm-producing isolates from sputum and respiratory secretions as compared to wounds (Piechota et al., 2018). This study also established a contrary relationship between biofilm from SSTIs ($P = 0.0036$) and sterile site infections ($P = 0.0281$) (Yang et al., 2017). Such variations in

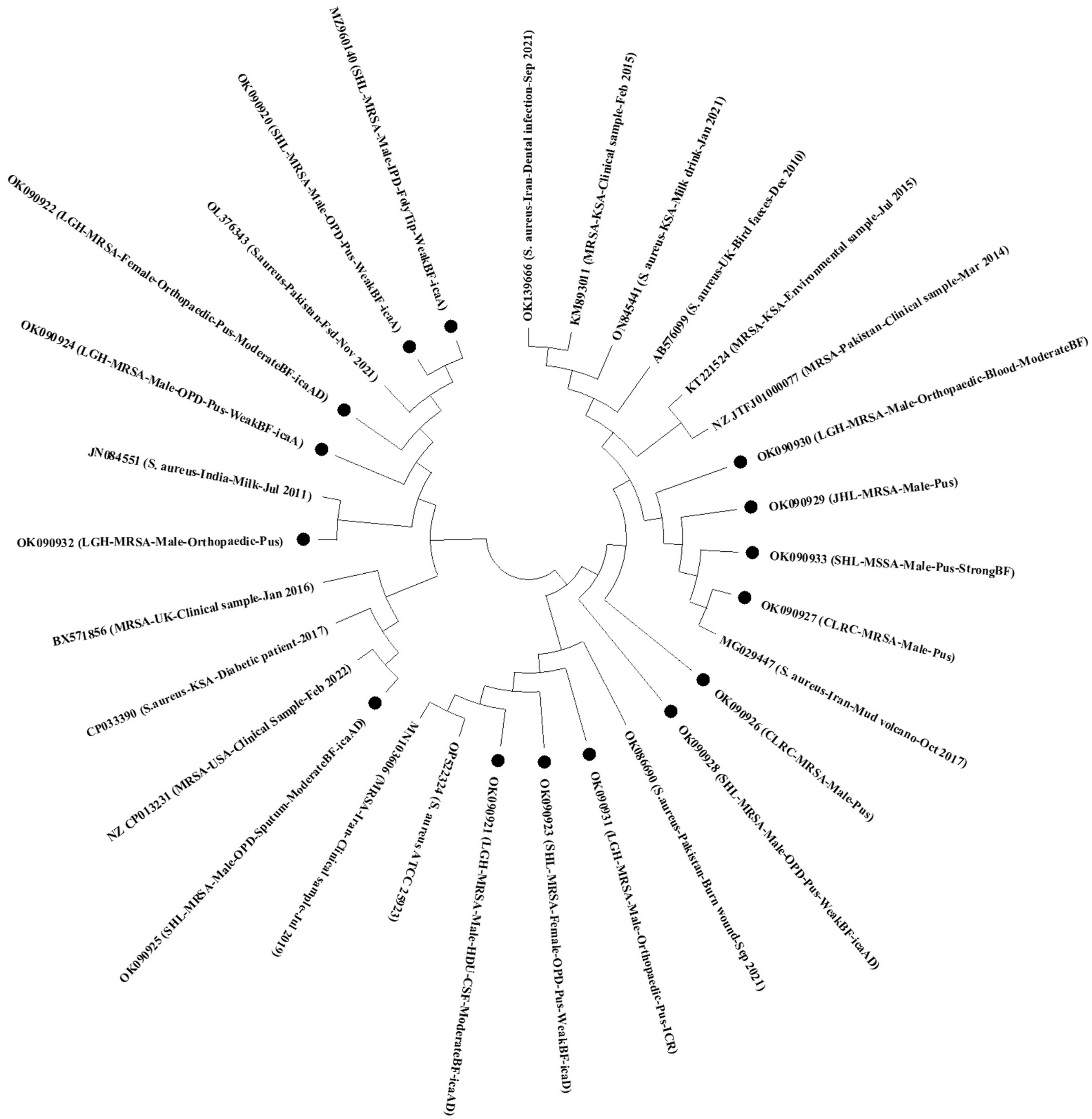

Fig. 3. Phylogenetic tree showing evolutionary relatedness of the representative 15 study isolates along with sequences from the literature. Phylogenetic analysis revealed the study isolated are clustered in two distinct clusters, the small cluster is sub-divided in two small sub-clusters. The study isolates are indicated by a • symbol.

biofilm formation across different infection sites underline the complexity of MRSA pathogenicity. Respiratory tract MRSA infections were scares in this country than reported in China and Iran (Samadi et al., 2017; Yang et al., 2017). Certain differences among the countries such as climate might contribute to these variations, for instance pneumonia is usually more prevalent in colder seasons. The blood and bone infections (1.33% each) and ICU admissions (4%) were also several-fold less than the previous study from China. These differences might be attributed to differences in targeted populations. The high prevalence of MRSA in wounds compared to urine aligns to a previous study from Pakistan (Khan et al., 2014).

The presence of *icaA* or *icaD* genes in the majority (92%) of the isolates supports the role of these genes in biofilm formation and similar findings are also reported in the literature (Piechota et al., 2018; Bazari et al., 2017). However, the study identifies isolates ($n = 4/143$, 2.8%) with *ica* genes that do not produce biofilm and vice versa ($n = 10/138$, 7.25%), indicating the existence of alternative mechanisms for biofilm formation which is also supported by previous studies (Vergara et al., 2017; Neopane et al., 2018; Cha et al., 2013). The specificity, sensitivity, positive predicted value (PPV) and negative predicted value (NPV) for the molecular assay were 100%, 95.83%, 100% and 50%, respectively. This indicates excellent performance of the molecular assay in identifying true positive and negative biofilm formers with a high reliability of the positive cases supported by a high PPV value. Since the biofilm detection rate by *ica* genes is similar to the slime layer detection assay, it is suggested that incorporation of the molecular assay for a rapid detection of biofilm-associated antibiotic resistance in MRSA isolates could be a potential solution for earlier accurate diagnosis of the infection.

The study establishes a significant association between biofilm formation and antibiotic resistance, emphasizing the challenges in treating biofilm-associated MRSA infections. Antibiotic resistance was higher in biofilm-forming (BF) isolates than in the non-biofilm formers (NBF), except for CPT, where no resistance was observed for any BF isolate. A significant association was observed for BF and antimicrobial resistance (AMR) in the case of CLR ($P < 0.01$), LEV ($P < 0.01$), SXT ($P < 0.01$) and FOS ($P < 0.01$). The previous study also found a high incidence of antimicrobial resistance in biofilms of *S. aureus* isolates and reported a strong association between biofilm and resistance in SXT, CIP, E and CN (Neopane et al., 2018). Our study also correlates with other previous studies (Samadi et al., 2017; Ou et al., 2020). In the current study, biofilm formation was also higher in strains which co-harbouring *icaAD* genes than those negative for the *ica* locus. Similar findings are also reported in another study (Manandhar et al., 2018). The biofilm has also resulted in multiple drug-resistant (MDR)-MRSA that accounted for 115/150 (76.7%) isolates, same to a previous study (Yang et al., 2017). The high prevalence of MDR-MRSA in this region underscores the limitations in antibiotic choices for empirical therapy. This observation highlights the need for a comprehensive approach to MRSA biofilm detection, considering multiple assays to adequately address empiric therapy for biofilm-producing antibiotic resistant isolates.

Phylogenetic analysis of the study isolates with representative isolates from Kingdom of Saudi Arabia (KSA), the US, the UK, Iran, India and some Pakistani isolates was performed (GenBank accession nos. mentioned in Fig. 3). It clustered the study isolates into two main clusters, the large and the small. The small cluster contained six study isolates and contained the weak and moderate biofilm forming isolates. This small cluster was further grouped into two small sub-clusters. They were similar to the samples from Pakistan, UK, USA, India and KSA. The large sub-group contained nine study isolates clustered closely with each other. They possessed similarities with the isolates from Iran, UK, KSA and a sample from the Islamabad region of Pakistan, suggesting that most of the isolates in the study location are confined locally and unique in the region. A further molecular investigation is warranted for further understanding the local strains dynamics.

5. Conclusions

The escalating antimicrobial resistance crisis in Pakistan is compounded by the high prevalence of biofilm-forming MRSA strains. This study, a pioneer in detecting *icaA* and *icaD* genes from diverse specimens across multiple healthcare centres in Pakistan, provides crucial insights into the biofilm landscape among MRSA isolates. The identified *icaAD* genes play a key role in defining biofilm-producing MRSA, influencing the strains' antibiotic resistance profiles. The high antibiotic resistance in biofilm-forming isolates poses significant challenges in treatment, potentially leading to reinfection and severe morbidity. Phylogenetic analysis indicates the uniqueness of MRSA strains in this region, necessitating enhanced molecular investigations to unravel the intricate mechanisms of pathogenesis. Understanding and mitigating the spread of such strains are imperative for alleviating the burden on healthcare settings.

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Ethical approval

Ethical approval for this study was obtained from the Ethical Committee of the CitiLab & Research Centre Lahore (15A - CLRC/30th).

CRediT authorship contribution statement

Asad Ali: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Saba Riaz:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Data availability

All the data is present in this research article.

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